



12492 - Confirming an 8th planet in the TRAPPIST-1 system

Cycle: 4, Proposal Category: DD

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JWST Proposal 12492 (Created: Wednesday, December 3, 2025, 3:00:25PM Eastern Standard Time) - Overview

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	TRAPPIST-1 b+e+i?	NIRSpec Bright Object Time Series	(1) TRAPPIST-1

ABSTRACT

The TRAPPIST-1 system now has an extensive set of GTO and GO observations with the James Webb Space Telescope, primarily targeting the atmospheric properties of the planets. We have used these datasets along with other transit times to detect an 8th planet with transit timing variations. There are three solutions which fit the data well, but based on dynamical arguments we favor a planet i which is near a 3:2 period ratio with planet h, giving a stable Laplace resonant angle between g, h and i which librates with small amplitude. This planet has a period of 28.7 days and a mass which is within a factor of 2 of the mass of Mars, and we expect it to lie in the same plane as the other seven planets, leading it to transit the star with a duration of 1.35 hours and depth of 0.1-2%. To date the putative transits of this planet have been missed, falling in a gap in the 20-day Spitzer observation, and falling below the signal-to-noise of a 75-day observation with K2. The planet has also been missed with prior ground-based and space-based observations, including JWST, thanks to its long orbital period. Much as the eighth planet in our Solar System, Neptune, was first discovered dynamically, and later confirmed photometrically, we wish to confirm this eighth planet photometrically by observing a transit. We ask for Director's Discretionary time to capture a transit with JWST using NIRSpec-PRISM, to confirm this planet, refine its properties, enable a density constraint, and to improve the modeling of the architecture and formation of the system. This would be the first exoplanet system with eight terrestrial transiting planets, twice as many as our Solar System.

OBSERVING DESCRIPTION

Our observations consist of a 18.61-hour time-series of TRAPPIST-1 aimed at detecting the transit of its eighth planet for the first time, discovered thanks to the transit-timing variation method. This window is our only opportunity before the star moves too close to the Sun. Fortunately, this transit window occurs close to an already scheduled TRAPPIST-1 observation as part of GO 9256. To optimize telescope time, we propose to extend the length of the already granted observation(GP Program 9256) using exactly the same instrument and setup: NIRSpec/PRISM with 5 groups per integration (1.377 seconds per integration) and the 512 subarray. In addition, NIRSpec/PRISM is the best mode to collect the maximum number of photons and maximize our chances of detecting the transit of TRAPPIST-1 i (expected depth ~1-2 ppt, while NIRSpec's white-light curve precision on the transit depth is ~90 ppm for a single transit). To maximize the probability of detection, the observation must take place between 2025-12-11 08:34:15 UT and 2025-12-12 09:34:15 UT, which includes 1 hour of scheduling flexibility to catch at least 1/2 of the transit of planet i at 3-sigma probability.

Proposal 12492 - Targets - Confirming an 8th planet in the TRAPPIST-1 system

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	TRAPPIST-1	RA: 23 06 30.3652 (346.6265217d) Dec: -05 02 36.70 (-5.04353d) Equinox: J2000	Proper Motion RA: 930.788 mas/yr Proper Motion Dec: -479.038 mas/yr Parallax: 0.0802123" Epoch of Position: 2016	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Taken from GTO 1331, updated based on Astrometry from Gaia DR3.</i> <i>Category=Star</i> <i>Description=[Exoplanet Systems, Exoplanets, M stars]</i> <i>Extended=NO</i>				

Proposal 12492 - Observation 1 - Confirming an 8th planet in the TRAPPIST-1 system

Observation	Proposal 12492, Observation 1: TRAPPIST-1 b+e+i?										Wed Dec 03 20:00:25 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRSpec Bright Object Time Series										
	Comments: We chose the start time as a 1-hour window with the duration of the observation selected to achieve a 3-sigma probability of catching at least 1/2 of a transit of planet TRAPPIST-1i under the assumptions that the 3:2 8-planet solution is correct, that the planet transits, and that the Laplace angle librates with an amplitude comparable to the inner 4 Laplace angle amplitudes with first-order resonances.										
Diagnostics	(TRAPPIST-1 b+e+i? (Obs 1)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(TRAPPIST-1 b+e+i? (Obs 1)) Warning (Form): Observers are responsible for checking that target acquisition is feasible.										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
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	Category=Star Description=[Exoplanet Systems, Exoplanets, M stars] Extended=NO										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	
Template	Subarray										
	SUB512										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	PRISM/CLEAR	NRSRAPID	5	24317	2	1	48634	66990.417		

Proposal 12492 - Observation 1 - Confirming an 8th planet in the TRAPPIST-1 system

Special Requirements	Between Dates 11-DEC-2025:09:34:15 and 11-DEC-2025:10:34:15 Time Series Observation No Parallel Attachments
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